Centrifugal Fans & Blowers



Fan & Blower Types

- Centrifugal Fans
- High Pressure Blower
- Medium Pressure Blower
- Low-Pressure Blower
- Axial Fans
- Plug Fans
- Direct Driven Fans
- Couple Driven Blower
- Belt Driven Blower
- Induced Draft (ID) Fan
- Forced Draft (FD) Fan
- High Temperature Fans
- Primary Fan
- Secondary Fan
- FRP (Fiberglass Reinforced Plastic) Fans
- Propeller Fans
- Tube Axial Fans
- All Types of Customised Industrial Fans, & Blowers

Leading Manufacturer of Blowers and Centrifugal Fans

We offer air pollution control equipment, specializing in high-quality blowers and centrifugal fans. Our advanced design capabilities, and experienced engineering team enable us to deliver tailored solutions that meet the unique needs of various industrial applications. Whether it's dust collection, ventilation, or process air handling, our durable and efficient fans ensure optimal performance while effectively addressing environmental challenges. With a focus on innovation, reliability, and cost-effectiveness.

Best Customized Solutions for Centrifugal Fans & Blowers

We specialize in providing the best customized solutions for centrifugal fans and blowers, tailored precisely to your industrial requirements. Our expertise covers a wide range of applications, from high-pressure ventilation to dust collection and process air handling.
Every project benefits from our advanced engineering capabilities, modern manufacturing processes, and a commitment to quality.

We understand that each industry has unique challenges, so we offer custom designs including special blade configurations, material selections, and performance adjustments to ensure optimal efficiency and reliability. Our centrifugal fans and blowers are built for durability and energy efficiency, making them ideal for demanding environments across sectors like power, steel, food processing, and more.

Industrial Applications

- Cement Plants
- Steel Plants
- Power Plants
- Incinerators
- Textile Industry
- Sugar Plants
- Paper Mills
- Coal Processing
- Fly Ash Management
- Pharmaceutical Sector
- Woodworking
- Material Recovery
- Dust collection
- Pneumatic conveying
- Scrubber exhaust
- Incineration
- Combustion air
- Pollution control
- Fume-hood exhaust
- Chemical process
- Dryer applications
- Bulk material handling
- Paper converting
- Other industrial applications.

CENTRIFUGAL AIR BLOWERS

Centrifugal Fans



SWSI & DWDI Configuration fans are designed for general HVAC and industrial applications where large volumes of clean air are required at low to moderate pressures. With backward curved non over loading type and forward curved design. Centrifugal fans offer the flexibility to match the performance and application at the highest efficiency.

Capacity : Up to 1,50.000 CMH,
Pressure: 10 to 150 mm WG.



Centrifugal Exhausters :

Fans are specifically designed for extraction or draft with duct loading. Useful for cupola furnaces. Also Useful for exhaust of moisture-laden, corrosive or chemically contaminated air. Same fans are available to meet any corrosive process requirements while maintaining quiet operation and high efficiency.

Capacity :Up to 50,000 CMH
Pressure: 50 to 550 mm WG. Temperature: Up to 660° C.

Medium & High Pressure Blowers :



We offer the most complete line of Pressure Blowers and high-capacity products in the market. These fans are used in applications where high pressures are required. The type of pressure blower selected will depend on the system requirements and temperature specifications. Some applications where pressure blowers are used include pneumatic conveying, combustion, and gas-handling, fuel firing, blowing, agitation, jet cooling systems.

Capacity: Up to 10,000 CMH,
Pressure: 500 to 1,500 mm WG.



Tube Axial Flow Fans :

Fans are designed for moderate-pressure ventilation and industrial-process applications where axial flow is desirable and space is at a premium. These fans are used for creating a high volume of airflow in lower-pressure applications. Also Uses include heat, smoke and fume removal, process- drying with ambient or preheated air, comfort and process cooling and general ventilation.

Capacity: 1,000 to 1,50,00 CMH
Pressure: 10 to 50 mm WG.

CENTRIFUGAL AIR BLOWERS

High Pressure Blowers :



Engineered for demanding applications, our High Pressure Blowers deliver powerful airflow at high static pressures, making them ideal for pneumatic conveying, combustion air supply, and industrial process ventilation. Constructed from robust mild steel or corrosion-resistant stainless steel, these blowers ensure reliable performance and long service life, even in challenging environments.

Air Flow: 100 – 30,000 CMH
RPM: 800 – 3,600
Material: Mild Steel, Stainless Steel

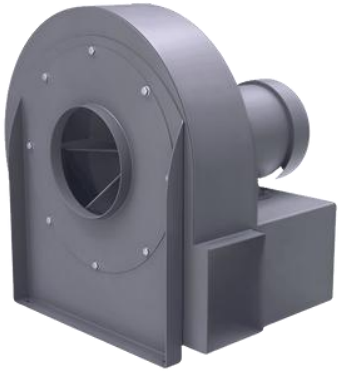


Low Pressure Blowers :

Designed for large-volume air movement at lower pressures, our Low Pressure Blowers are perfect for ventilation, air handling, and general exhaust systems. With a wide range of capacities and durable construction from carbon or alloy steel, these blowers provide efficient and economical solutions for factories, warehouses, and process industries.

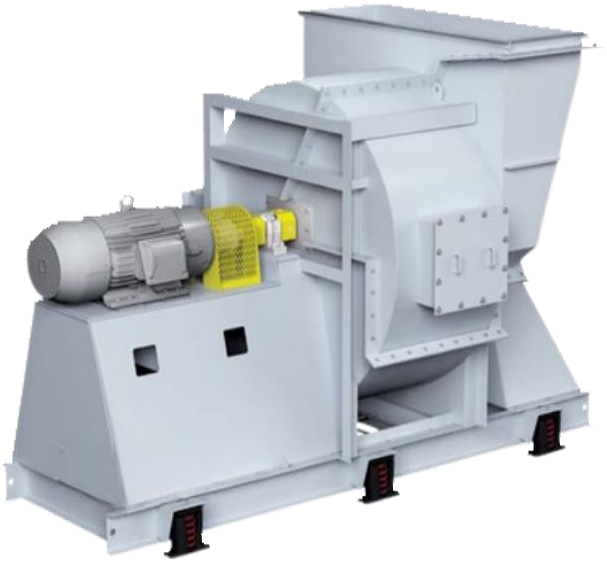
Air Flow: 1,000 – 300,000 CMH
RPM: 500 – 1,450
Material: Carbon Steel, Alloy Steel

Medium Pressure Blowers :



Our Medium Pressure Blowers offer a balanced solution for applications requiring moderate airflow and pressure. Suitable for dust collection, material handling, and process air supply, these blowers are built from high-quality carbon or alloy steel to ensure consistent performance and durability in diverse industrial settings.

Air Flow: 300 – 300,000 CMH
RPM: 800 – 3,000
Material: Carbon Steel, Alloy Steel



Customised Centrifugal Fans & Blowers :

specializes in fully customized centrifugal fans and blowers, engineered to meet your specific operational requirements. Whether you need unique airflow rates, special materials, or tailored configurations, our expert team delivers solutions that optimize efficiency, reliability, and performance for your application.

Air Flow: 1,000 – 500,000 CMH
RPM: 800 – 3,000
Material: Carbon Steel, Alloy Steel

Plug Fan :

Ideal for high-temperature and recirculation applications, our Centrifugal Plug Fans are designed to handle airflows at temperatures up to 300°C. Constructed from premium carbon or alloy steel, these fans are perfect for ovens, dryers, and process heating systems, ensuring safe and efficient operation under extreme conditions.

Air Flow: 1,000 – 150,000 CMH
Operating Temperature: Up to 300°C
Material: Carbon Steel, Alloy Steel





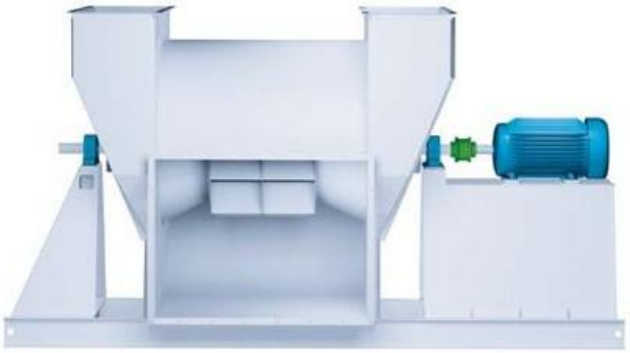
Centrifugal DWDI Fan (Double Width Double Inlet) :

Our Centrifugal DWDI Fans feature a double-width, double-inlet design for maximum airflow and efficiency in large-scale ventilation and air handling systems. Built from robust carbon or alloy steel, these fans are ideal for HVAC, industrial ventilation, and air pollution control applications requiring high volume and dependable performance.

- Air Flow – 1000 CMH to 5,00,000 CMH
- Static Pressure – 50 mmWC to 200 mmWC
- Motor Power – up to 250 kW
- RPM – 400 to 1000
- Material of Construction – Carbon Steel, Alloy Steel, Stainless Steel, Weldox Steel, Duplex Steel, FRP & Rubber Lined.
- Types of Drive – Direct / V-Belt



Belt Driven Extended Structural Frame to Mount Motor



Direct Drive - Double Width Fan with Integral (Attached) Inlet Box Motor Coupled to Fan Shaft (common fan base included)

High-Performance Axial Fans :

Axial Fans are expertly engineered to deliver superior air movement and optimal performance in demanding industrial environments. Our axial fans are the ideal choice for efficient ventilation and cooling, ensuring a healthy and comfortable workspace while supporting your operational needs.

Available in a wide range of configurations—including tube axial, wall-mounted, and square duct types our fans are designed to address diverse application requirements. Each model is constructed with precision using high-quality materials, ensuring durability, reliability, and long service life even under continuous operation.



Direct Driven



Belt Driven



Tubular Type Fans



Direct Drive - Swingout Construction

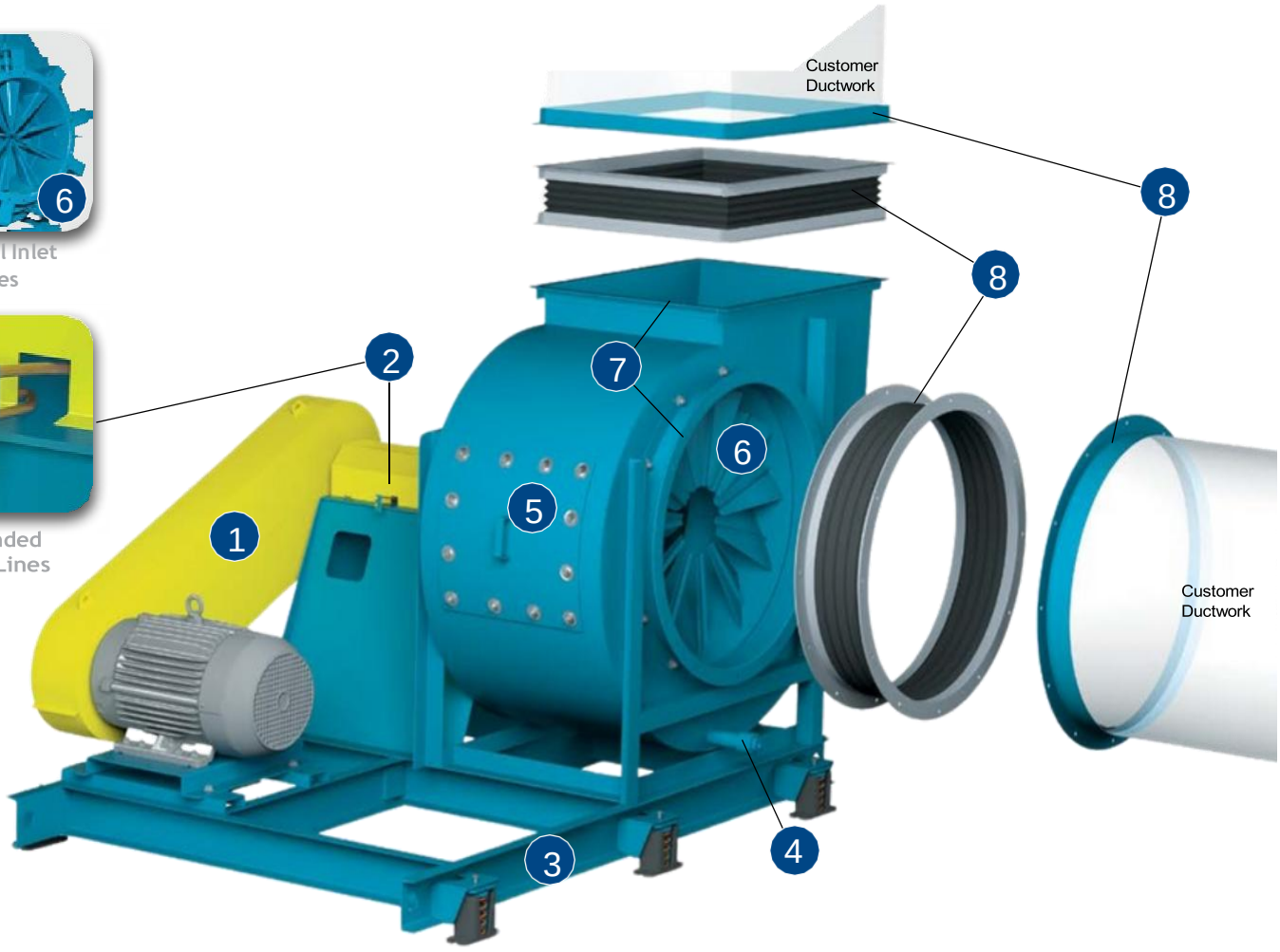
Fan Components



External Inlet Vanes



Extended Lube Lines



1. **Belt Guard:** Belt guard protects personnel from the moving drive parts.
2. **Shaft Guard, Bearing Guard & Extended Lube Lines** Sheet metal guards cover shaft and bearings and come with extended lube lines to a common point outside of the guard. A guard spanning the shaft between the bearings is available to provide open access to bearings for lubrication and vibration monitoring.
3. **Vibration Isolation Bases** provide a common support to fan, motor and drive including guards and utilize heavy duty structural channel. Vibration isolation bases require spring or rubber-in-shear type isolators that are designed to limit forces transmitted to the support structure of an operating fan. Flexible connectors at inlet and outlet are also required (as shown).
4. **Drain with Plug** Threaded pipe coupling welded to the lowest point in the housing scroll. All fans come with a weep hole in the bottom of the housing as standard.
5. **Bolted Access Doors** for wheel inspection and maintenance.
6. **Nested & External Inlet Vanes** Variable inlet vanes cause the entering air to spin in the direction of wheel rotation, resulting in reduction in volume, static pressure and brake horsepower and thus providing an infinite number of fan curves approximately parallel to the original fan curve. Variable inlet vanes cost about 50% to 80% more than outlet dampers but offer significant savings in energy. Both types are available to 315°C construction.
7. **Flanged Inlets/Outlets** Punched inlet flanges are available for duct mounting (as shown). Punched or unpunched outlet flanges are welded to the fan outlet. Unpunched flanged outlet is standard on all SWSI and DWDI Class III and IV fans.
8. **Inlet/Outlet Companion Flanges & Flex Connectors** (Round & Rectangular) Companion flanges are commonly connected to a user's duct for easy installation of flexible connections between the fan and duct. Companion flanges and flex connectors are punched to match the fan's inlet or outlet punching.

Fan Components



9 **Inlet & Outlet Safety Screens** are available for mounting in the fan inlet or outlet in non-ducted applications.

10 **Outlet Dampers (Opposed or Parallel Blade)** Outlet dampers add resistance to the fan by shifting the operating point to the left of the rating point. The horsepower savings depends on the relative position on the fan curve and is usually much less than other methods. Outlet dampers are typically the least expensive option and should be considered when infrequent operation at lesser capacity is desired or when handling hot, humid or particulate laden air. **Parallel blade dampers** are recommended for systems where air volume is modulated between full-open to about 75% of open. **Opposed blade dampers** cost about 10% more and are recommended for systems where volume is modulated over the entire range. Opposed blades reduce air volume in a closer relationship to the control arm movement. Both types are available to 400°C construction.

11 Unitary Base utilize structural channel to support the fan assembly and are designed for use without isolators.

High Temperature Construction
Standard fan design options are available to handle airstream temperatures to 425°c. Consult Fan & Blower representative for applications over 425°C. The fan bearings should be kept outside of the hot airstream and below 50°C ambient.

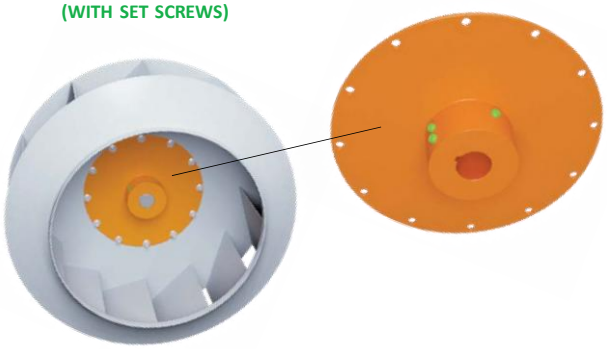


Shaft Cooler & Safety Guard

12 **Shaft Cooler (Heat Slinger)**
Cast aluminum shaft cooler dissipates the heat transferred to the shaft from the airstream protecting the fan bearings. Recommended for applications over 148°C.

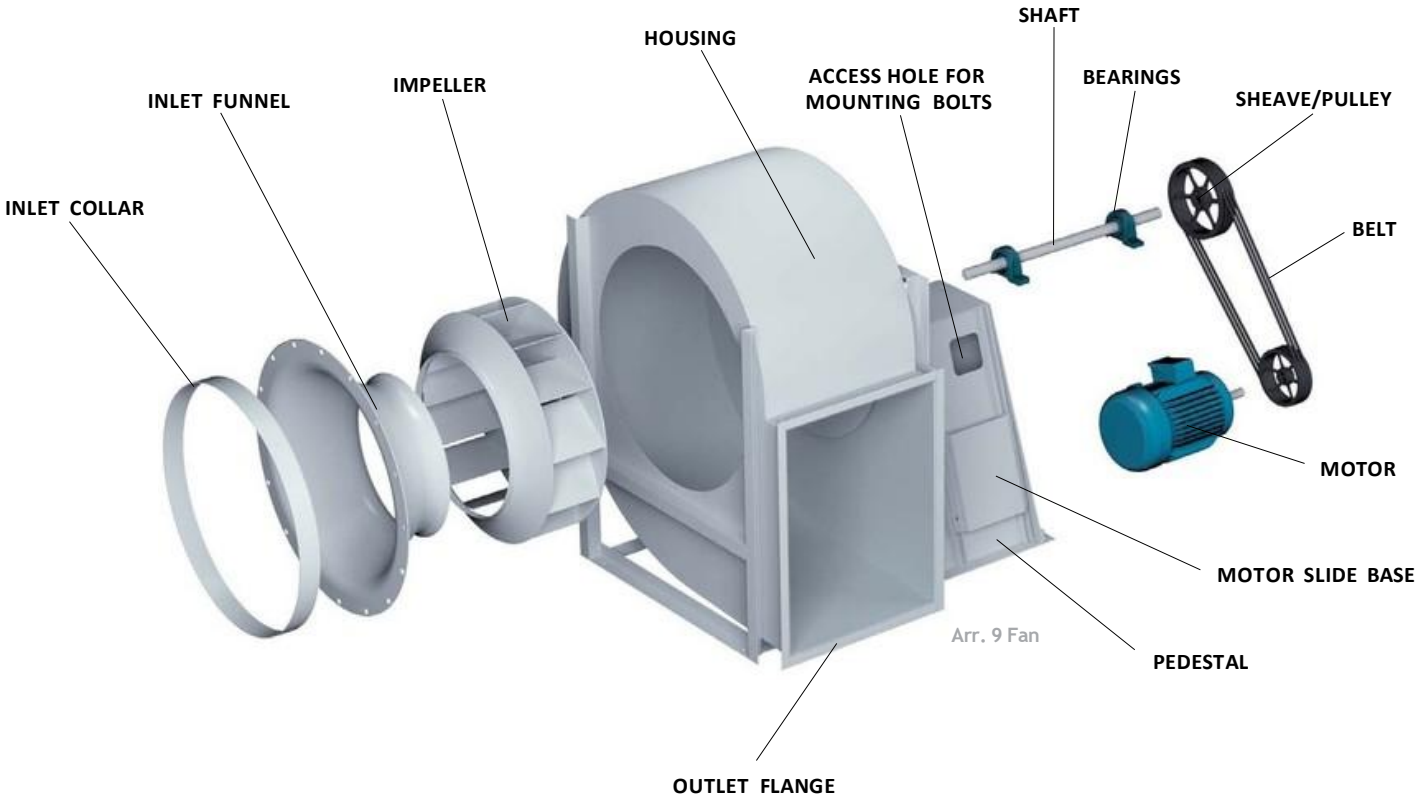
13 **STRAIGHT BORE HUB**
The bore of the hub is straight through. Shafts are keyed and mounted to the hub.

STRAIGHT BORE HUB
(WITH SET SCREWS)



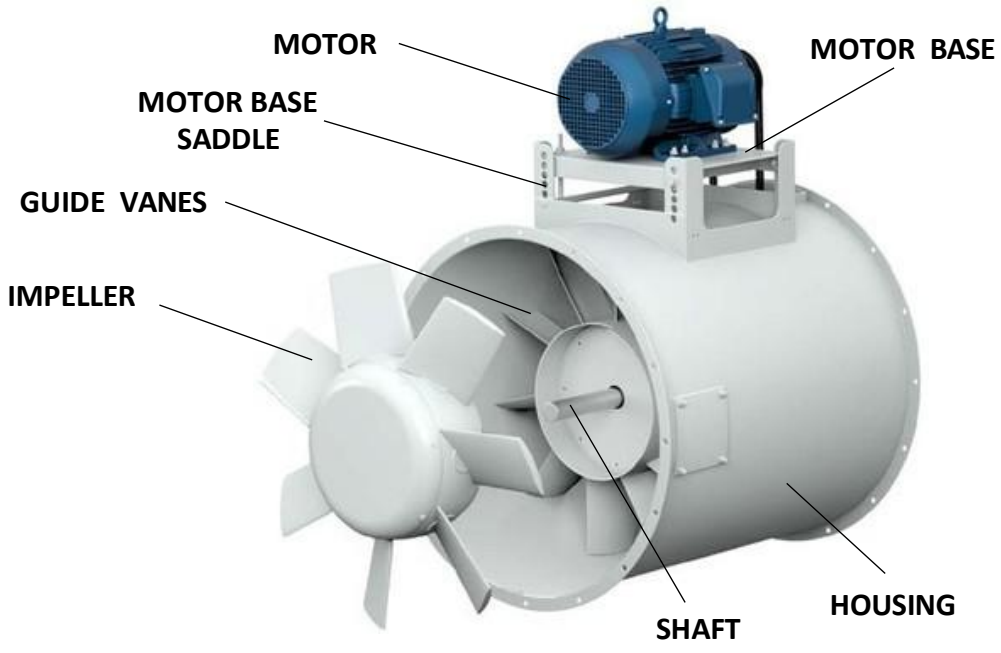
Fan Components

CENTRIFUGAL FANS & BLOWERS

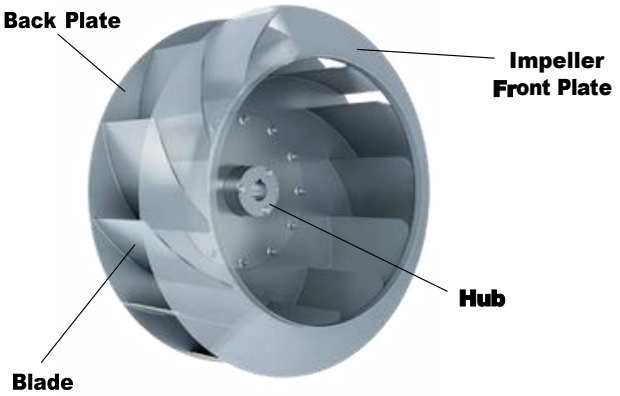


Arr. 9 Fan

BELT DRIVEN VANEAXIAL FANS



IMPELLER TYPES CENTRIFUGAL



Backward Curved

Impeller with single thickness,
backward curved blades



Backward Inclined

Impeller with flat, single thickness,
backward inclined blades



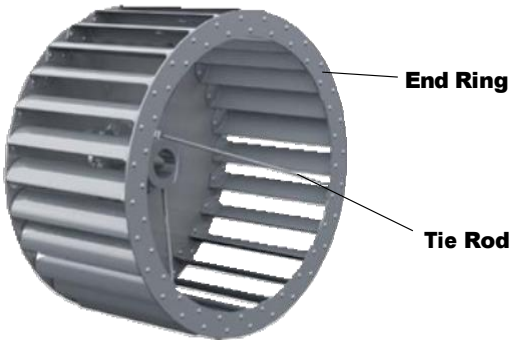
Airfoil

Impeller with airfoil,
backward inclined blades



Radial Tip

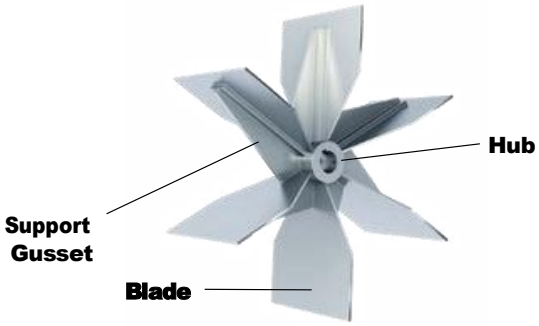
Impeller with blade design curved forward at the
entering edge and radial at the tip of the leaving edge



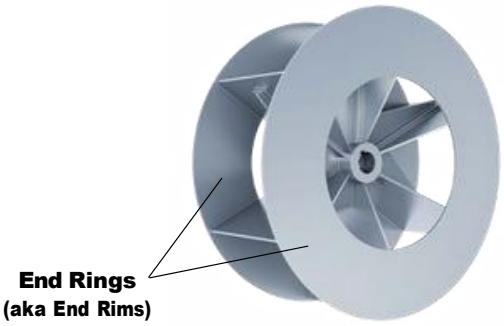
Forward Curved

Impeller with single thickness,
forward curved blades

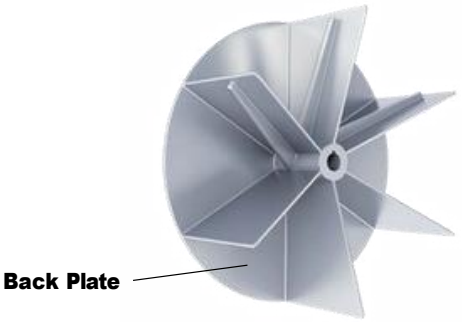
IMPELLER TYPES CENTRIFUGAL



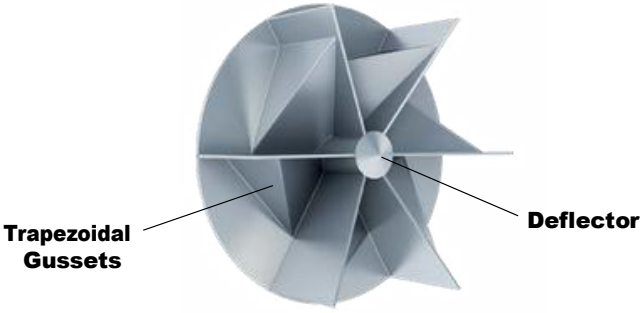
Radial Bladed
Paddle Impeller (Open Type)
Impeller with single thickness, radial paddle type blades



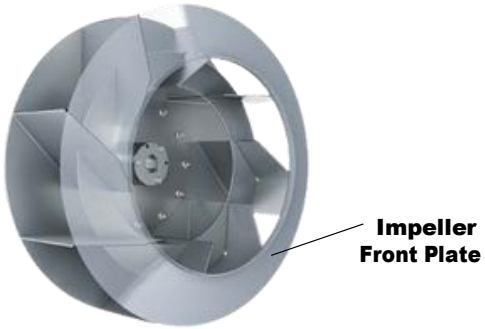
Radial Bladed
Paddle Impeller
Similar to the open type radial impeller design, except with the addition of front and back end rings



Radial Bladed
Material Handling Impeller (Wool)
Similar to the open type radial impeller design, except with a full back plate



Radial Bladed
Paper Handling Impeller
Constructed with full back plate gussets for extra rugged durability



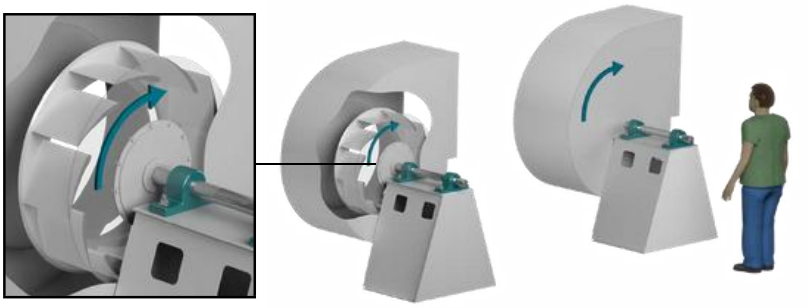
Radial Bladed
Air Handling Impeller
Constructed with heavy-gauge blades welded to both back plate and impeller front plate



Mixed Flow
Airfoil blades taper from impeller front plate to hub end resulting in a larger tip width

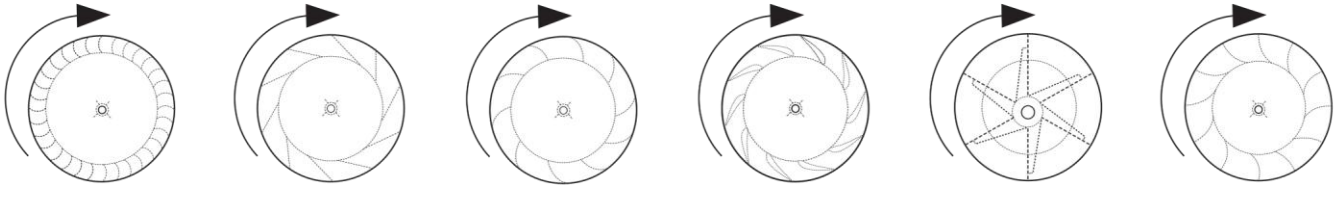
DISCHARGES & IMPELLER ROTATION

SWSI CENTRIFUGAL FANS



CLOCKWISE (CW)

ROTATION VIEW **FROM DRIVE SIDE**



Forward Curved | **Backward Inclined** | **Backward Curved** | **Airfoil** | **Radial Bladed** | **Radial Tip**



UBD
Upblast
CW 360
Also known as:
- Clockwise 360°

TAU
Top Angular Up
CW 45
Also known as:
- Clockwise 45°

THD
Top Horizontal
CW 90
Also known as:
- Clockwise 90°

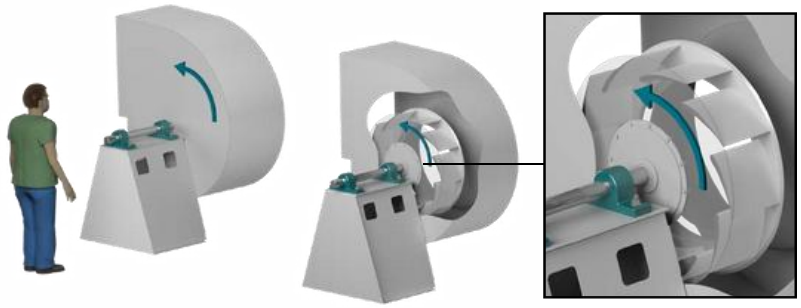
TAD
Top Angular Down
CW 135
Also known as:
- Clockwise 135°

DBD
Downblast
CW 180
Also known as:
- Clockwise 180°

BAD
Bottom Angular Down
CW 225
Also known as:
- Clockwise 225°

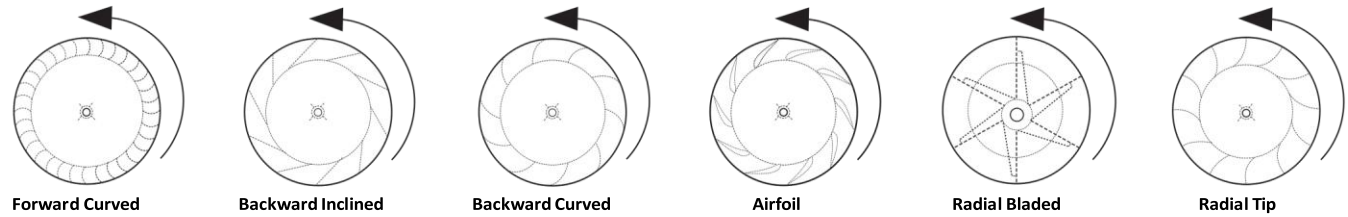
BHD
Bottom Horizontal
CW 270
Also known as:
- Clockwise 270°

BAU
Bottom Angular Up
CW 315
Also known as:
- Clockwise 315°



COUNTERCLOCKWISE (CCW)

ROTATION VIEW **FROM DRIVE SIDE**



Forward Curved | **Backward Inclined** | **Backward Curved** | **Airfoil** | **Radial Bladed** | **Radial Tip**



UBD
Upblast
CCW 360
Also known as:
- Counterclockwise 360°

TAU
Top Angular Up
CCW 45
Also known as:
- Counterclockwise 45°

THD
Top Horizontal
CCW 90
Also known as:
- Counterclockwise 90°

TAD
Top Angular Down
CCW 135
Also known as:
- Counterclockwise 135°

DBD
Downblast
CCW 180
Also known as:
- Counterclockwise 80°

BAD
Bottom Angular Down
CCW 225
Also known as:
- Counterclockwise 225°

BHD
Bottom Horizontal
CCW 270
Also known as:
- Counterclockwise 270°

BAU
Bottom Angular Up
CCW 315
Also known as:
- Counterclockwise 315°

ARRANGEMENTS - CENTRIFUGAL FANS



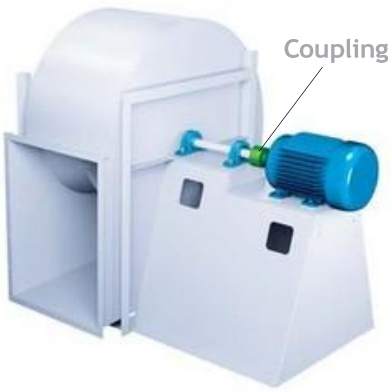
Direct Driven

Direct Drive Impeller Mounted to Motor Shaft



Belt Driven

Belt Driven Extended Angle **F**rame to Mount Motor



Couple Driven

Direct Drive Motor Coupled to Fan Shaft



Flange Mount Blower



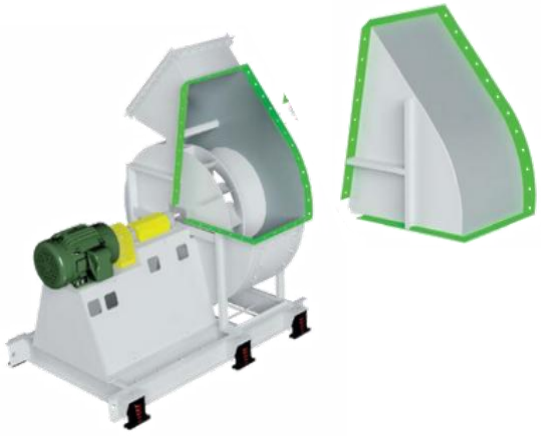
Flange with Foot Mounted

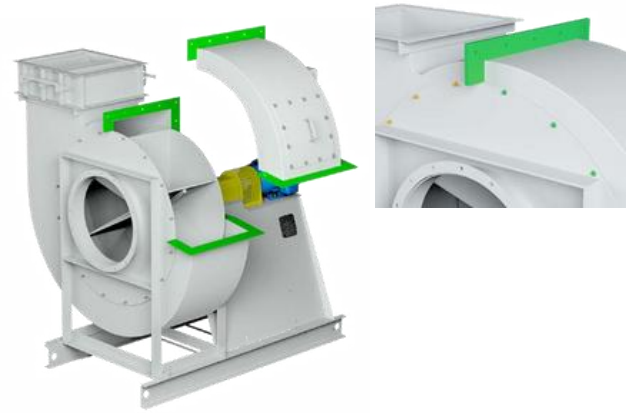


Belt Driven Motor Mounted Inside of Pedestal

SPLIT HOUSINGS

(Typical for impeller removal) Housings are split at angles 90 degrees or greater to facilitate impeller removal without disturbing inlet or outlet.

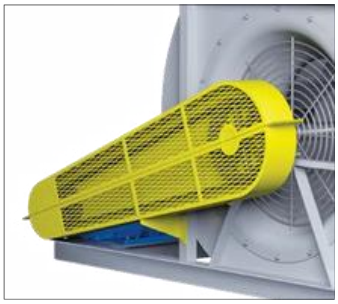




Accessories



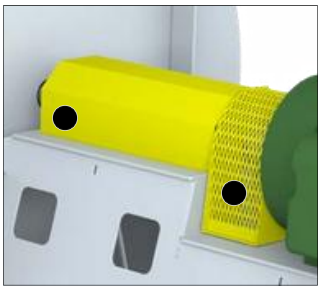
Inspection window



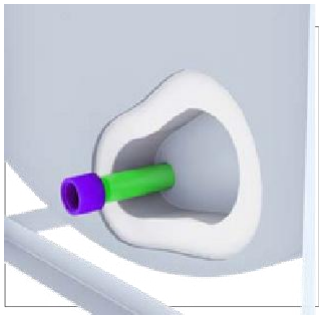
BELT GUARD



BLAST GATE



SHAFT/BEARING GUARD
COUPLING GUARD



DRAIN



EVASÉ



FINS ON IMPELLER

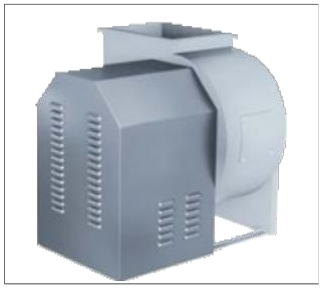


FLANGES (Inlet and
Outlet) (Round and
Rectangular)



INLET/OUTLET
FLEX CONNECTORS
(Round and Rectangular)
Also known as: - Expansion Joint

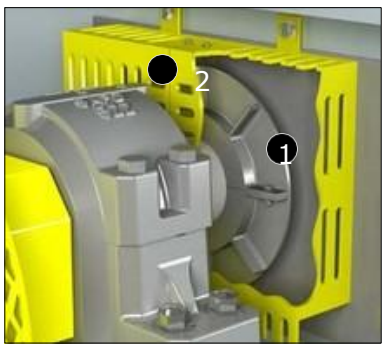




MOTOR COVER / WEATHER COVER



VIBRATION Pads



SHAFT COOLER &
COOLER GUARD



SILENCER
(with support legs)

Silencers are available for bot
h the inlet and outlet of fans



RECTANGULAR EVASÉ



ROUND EVASÉ